Convergent Divergent Nozzle Design

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This project focuses on the design, manufacturing, and performance of a Converging-Diverging nozzle. The objective of this lab was to characterize the internal flow conditions under choked conditions by using multiple different methodologies. A converging-diverging nozzle was designed under specific constraints with a maximum length of 6 inches and a throat to pitot tube area ratio of less than 10%. Using Ansys CFD we evaluated the performance of the nozzle and physically tested it at the North Carolina State University wind tunnel.

Nomenclature

a	=	speed of sound
A	=	area along nozzle
A^*	=	area at nozzle exit
$\dot{m}$	=	mass flow rate
M	=	Mach number
P	=	static pressure
P_0	=	stagnation pressure
P_{01}	=	tank total pressure
P_{02}	=	pitot total pressure
P_s	=	pitot static pressure
R	=	ideal gas constant, 287 $\frac{J}{kg*k}$
T	=	statictemperature
T_0	=	stagnation temperature
T_{01}	=	tank temperature
T_{02}	=	pitot temperature
V	=	flow velocity
ρ	=	static density
ρ_0	=	stagnation density
γ	=	ratio of specific heats, 1.4

I. Introduction

The study of compressible flow through nozzles is the most common aspect used in aerospace engineering, these studies are critical for the development of propulsion systems and many different high speed aerodynamic applications. The primary goal of this lab was to design and manufacture a functional CD nozzle. We employed three different approaches to ensure a valid comparison: analytical calculation where we used 1D isentropic flow to predict flow properties and choked conditions, CFD where we used ANSYS software to model and visualize pressure and velocity distributions, and finally experimental testing where we measured actual flow data using a pitot tube set up in the wind tunnel.

II. Methodology

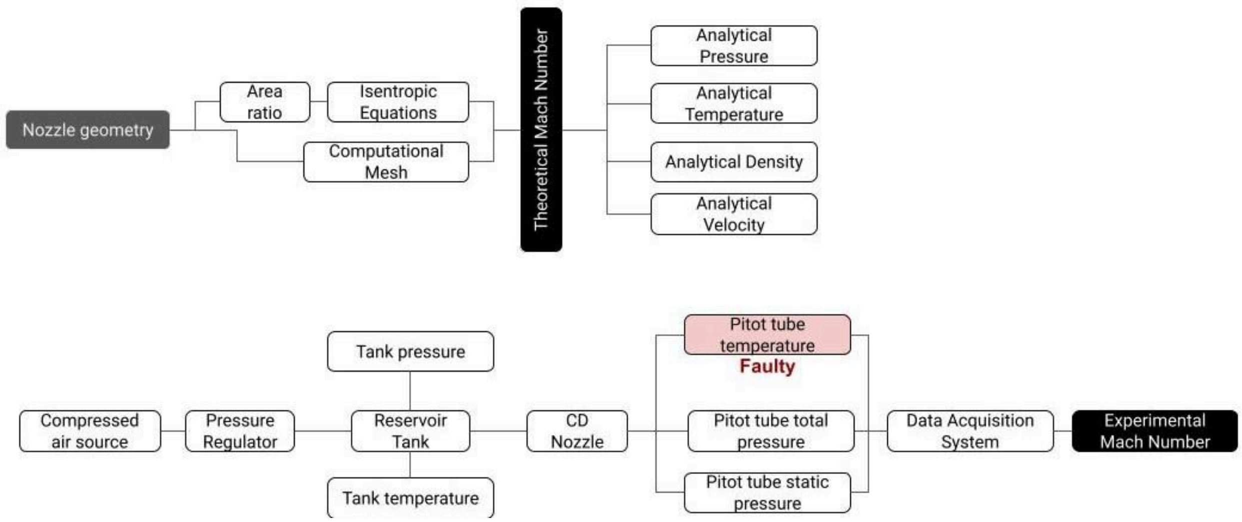

Figure 1: Experimental Flow diagram

In this experiment we designed, manufactured and tested a CD nozzle to analyze supersonic flow characteristics. Using the specific constraints the nozzle was created and manufactured using a 3D printer. Analytical calculations were performed to verify that the required total pressure didn't exceed the 80psi limit that we put as well as the throat area designed to maintain a Pitot tube blockage ratio of less than 10%. With analytical calculations using the equations below, a Mach number of 1.84 was reached and an exit radius to throat radius ratio was calculated at 1.234. Testing was done in North Carolina State University's wind tunnel where a flow was driven by pressure and adjustable using a regulator. A pitot tube was mounted on a traversable mount and moved through the nozzle at increments of .5in and 1in.

The nozzle was designed in Solidworks following the given constraints for geometry listed in the Lab 5 handout. Listed below are the pictures of the 3D printed final design which was created with PLA+ filament. One picture shows the entire revolved section of the nozzle while the other demonstrates the half section in order to clearly see the internal converging and diverging sections. Also listed below is the 3D sketch model on Solidworks which labels each geometric dimension used for production and analytical calculations.

A. Analytical analysis

$$\frac{A}{A^*} = \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \frac{\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}}{M} \tag{1}$$

$$\frac{p}{p_0} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma}{\gamma-1}} \quad \textbf{(2)}$$

$$\frac{p_s}{p_{02}} = \left(\frac{\gamma+1}{1-\gamma+2\gamma M^2}\right)\left[\frac{(\gamma+1)^2M^2}{4\gamma M^2-2(\gamma-1)}\right]^{-\frac{\gamma}{\gamma-1}} \quad \textbf{(3)}$$

$$\frac{T}{T_0} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{-1} \quad \textbf{(4)}$$

$$\frac{\rho}{\rho_0} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{-\frac{1}{\gamma-1}} \quad \textbf{(5)}$$

$$a = \sqrt{\gamma RT} \quad \textbf{(6)}$$

$$V = Ma \quad \textbf{(7)}$$

$$\dot{m} = \rho AV \quad \textbf{(8)}$$

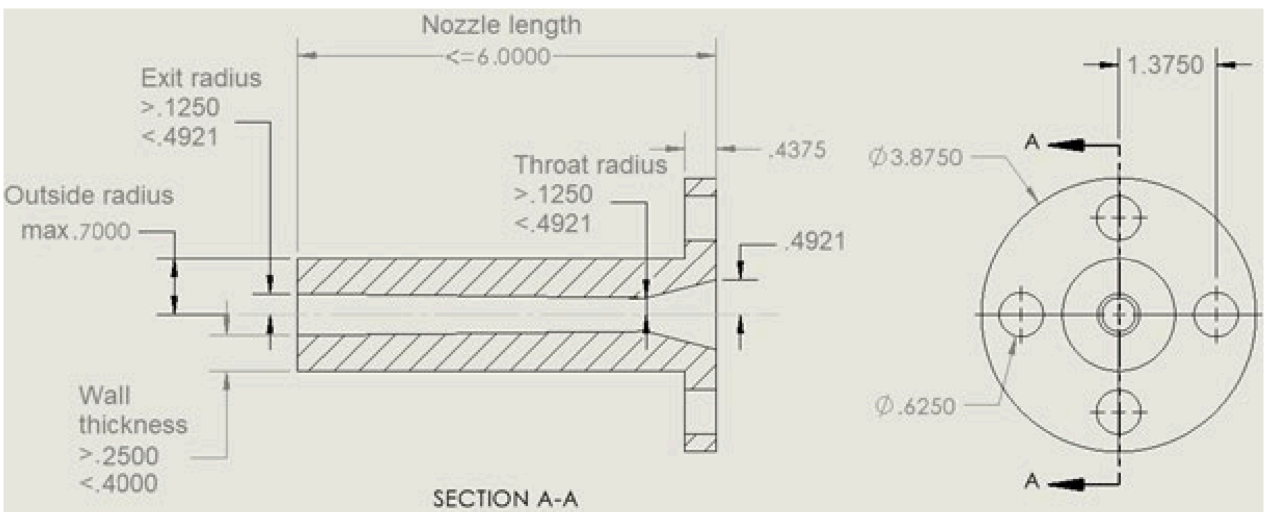

Figure 2. Nozzle design constraints

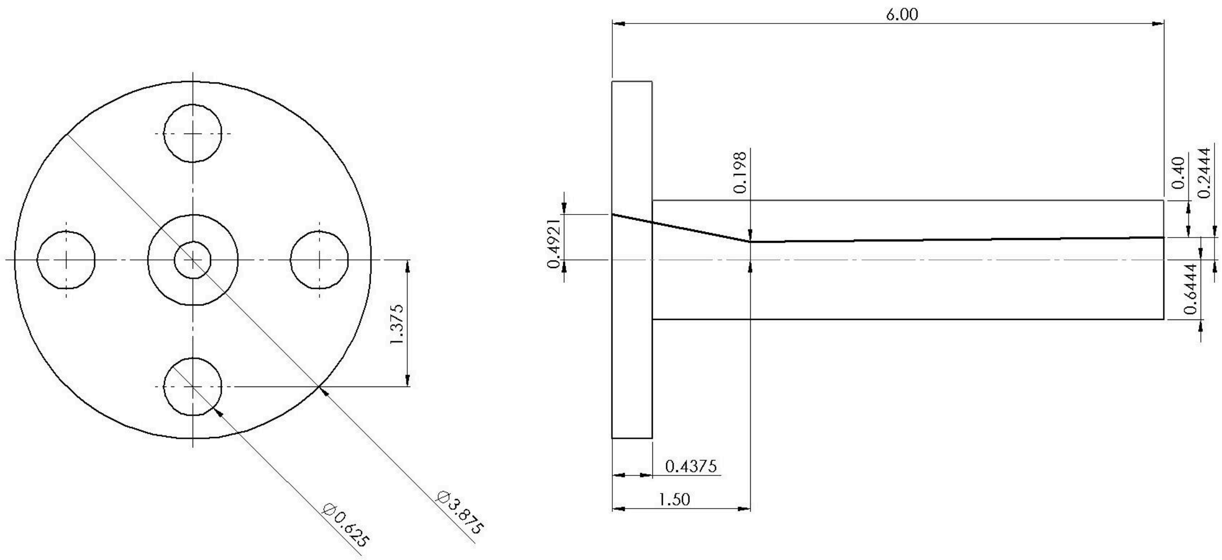

Figure 3. SolidWorks nozzle design

Given the known geometry of the converging nozzle from fig (3), the area ratio can be used to calculate the Mach number using Eq(1). From there, the Mach number and the static conditions can give the pressure, temperature, and density of the flow using isentropic Eq (2), Eq (3), and Eq (4) respectively. With the flow temperature, the speed of sound can be calculated with Eq (6), which in turn gives velocity when combined with the Mach number using Eq (7). Finally, the flow velocity, density, and nozzle area can be combined to find the mass flow rate with Eq (8).

B. Experimental analysis

The experimental analysis differs from the analytical analysis in that the pressure ratio is known instead of the nozzle geometry. As such, Eq (2) is used to calculate the experimental Mach number in the converging section where the flow is subsonic, and the Rayleigh pitot formula in the diverging section where the flow is supersonic. The temperature data comes from the tank and pitot tube readings.

III. Results

A. Analytical analysis

Table 1. Data acquired from isentropic equations

R (J/K.kg)		γ		p_0 (Pa)		T_0 (K)		ρ_0 (Kg/m^3)	
287		1.4		482633		300		6.4063	
Point	**Location (mm)**	**Area (mm^2)**	**Pressure (Pa)**	**Temperature (K)**	**Density (kg/m^3)**	**Velocity (m/s)**	**Mach number**	**Mass flow (kg/s)**	**Note**
1	0	486.6	479471	299.44	5.579	33.6	0.10	.0913	Inlet
2	12.7	313.6	474938	298.63	5.542	52.5	0.15	.0913	
3	25.4	178.4	457905	295.53	5.399	94.8	0.28	.0913	
4	38.1	81.1	254966	250.00	3.554	316.9	1.00	.0913	Throat
5	50.8	84.72	187595	229.02	2.854	377.6	1.24	.0913	
6	63.5	88.44	161793	219.54	2.568	402.1	1.35	.0913	
7	76.2	92.24	143351	212.07	2.355	420.3	1.44	.0913	
8	101.6	100.1	116915	200.08	2.036	448.0	1.58	.0913	
9	127	108.3	98139	190.31	1.797	469.4	1.70	.0913	
10	152.4	116.7	83877	181.96	1.606	487.0	1.80	.0913	Exit

Table 1 shows all of the values derived during the analytical analysis. As distance down the nozzle increases, Mach number and velocity increase, while pressure, temperature, and density decrease and mass flow rate stays the same. All the values except mass flow rate have a sharp rate of change at 25 mm along the nozzle, which corresponds to the nozzle throat. This aligns with the theoretical expectations of a choked flow at the throat.

One thing to consider is that the analytical analysis predicts the flow at the nozzle exit will be overexpanded. The pressure at the nozzle exit is predicted to be 83,877 Pa, while atmospheric pressure is 101,325 Pa. This means the nozzle may create shock diamonds, and there is some efficiency loss compared to a perfectly expanded flow.

B. Ansys simulation

As shown in table 1 the experimental mach number generally decreased as the pitot tube moved further into the nozzle. The airflow measured started supersonic and then decreased as the pitot tube moved into the converging diverging nozzle. This generally matches the analytical results and shows a strong direct correlation between mach number and the distance travelled by the airflow in the nozzle.

Ansys was used for CFD analysis of our designed geometry for the CD nozzle. Listed below are the results we obtained from the video guide listed in the lab 5 handout. With an expected inlet/tank pressure of 80 psi, the following results shown in order are the Mach number contours, Static Pressure contours, and finally the Static

Temperature throughout the nozzle. Our solutions follow the standard results of CD nozzle behavior by going from subsonic to supersonic Mach numbers, high to low static pressures, and high to low static temperatures from inlet to exit. As the air gets compressed through the converging section to expanded from the diverging section, the solutions follow the typical trends for such geometry.

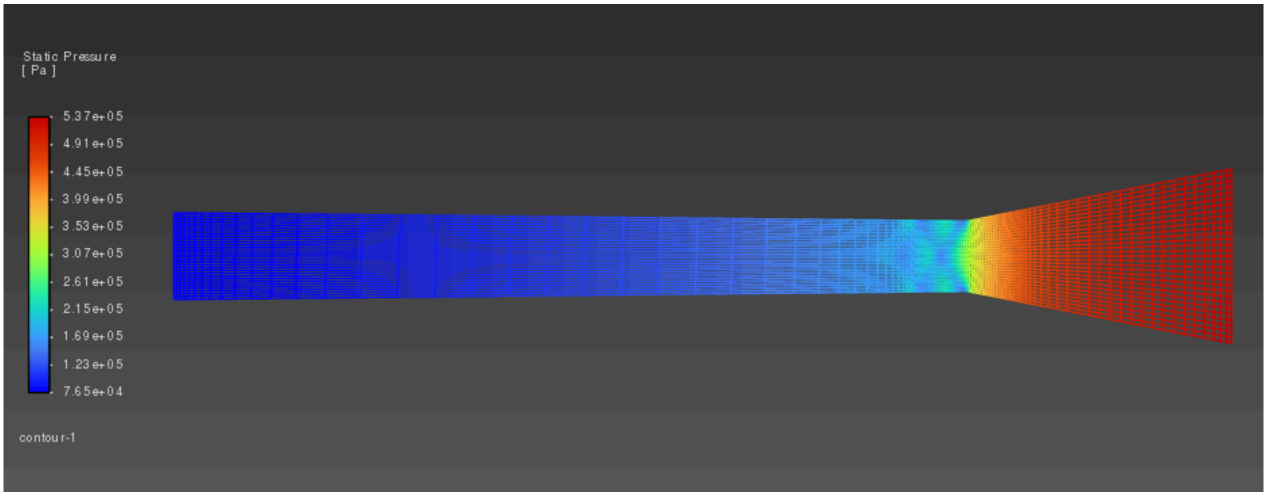

Figure 4: Ansys static pressure

As seen in the contour above, Statice Pressure reached a maximum value of 5.37e+05Pa at the inlet, and a minimum of 7.65e+04Pa at the outlet.

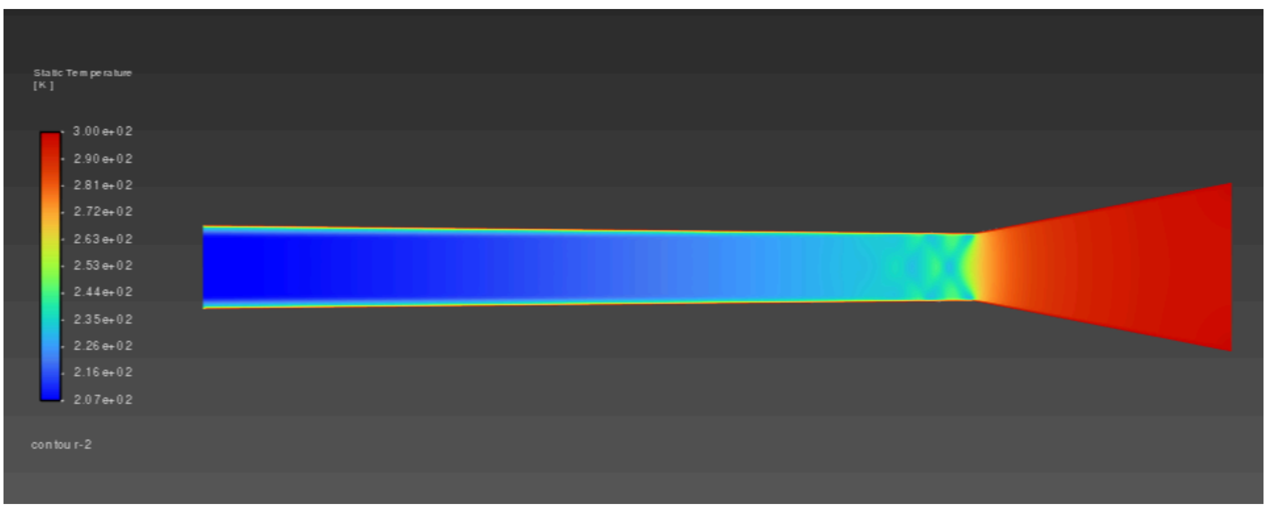

Figure 5: Ansys static temperature

As seen in the contour above, Static Temperature reached a maximum value of 3.00e+02K at the inlet, and a minimum of 2.07e+02K at the outlet.

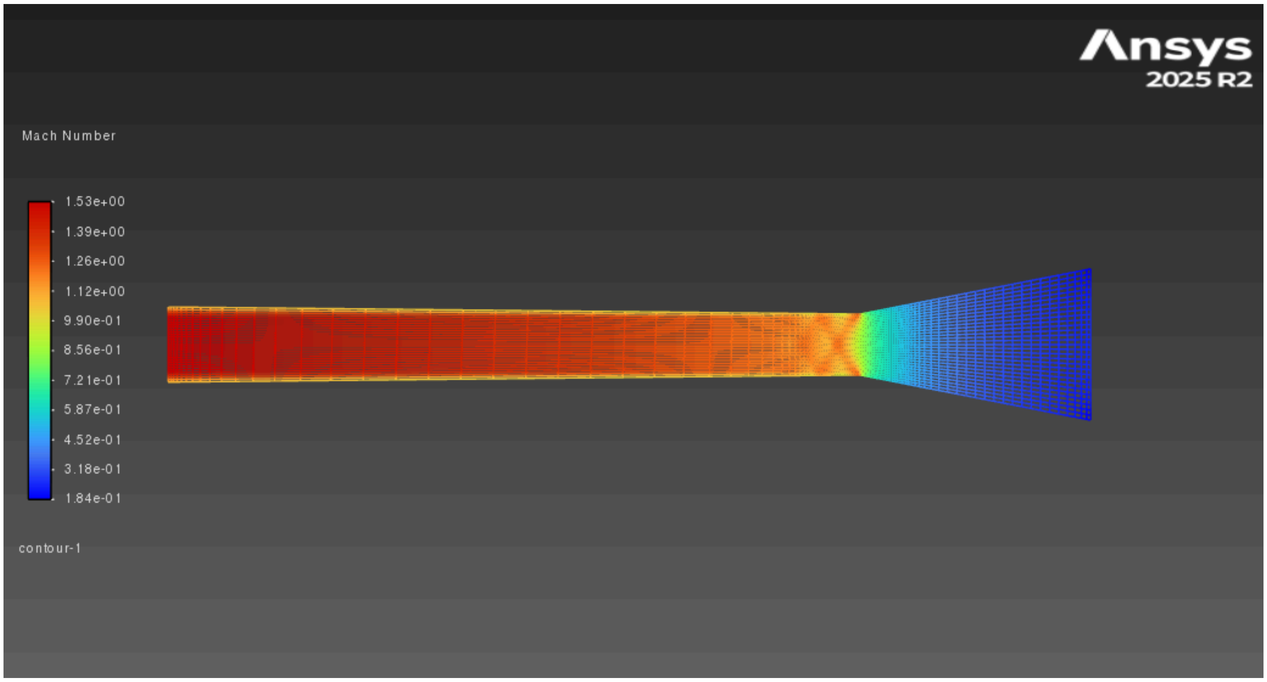

Figure 6: Ansys Mach number

As seen in the contour above, the Mach number reached a minimum value of 0.184 at the inlet, and a maximum value of 1.53 at the outlet.

C. **Mach number along centerline**

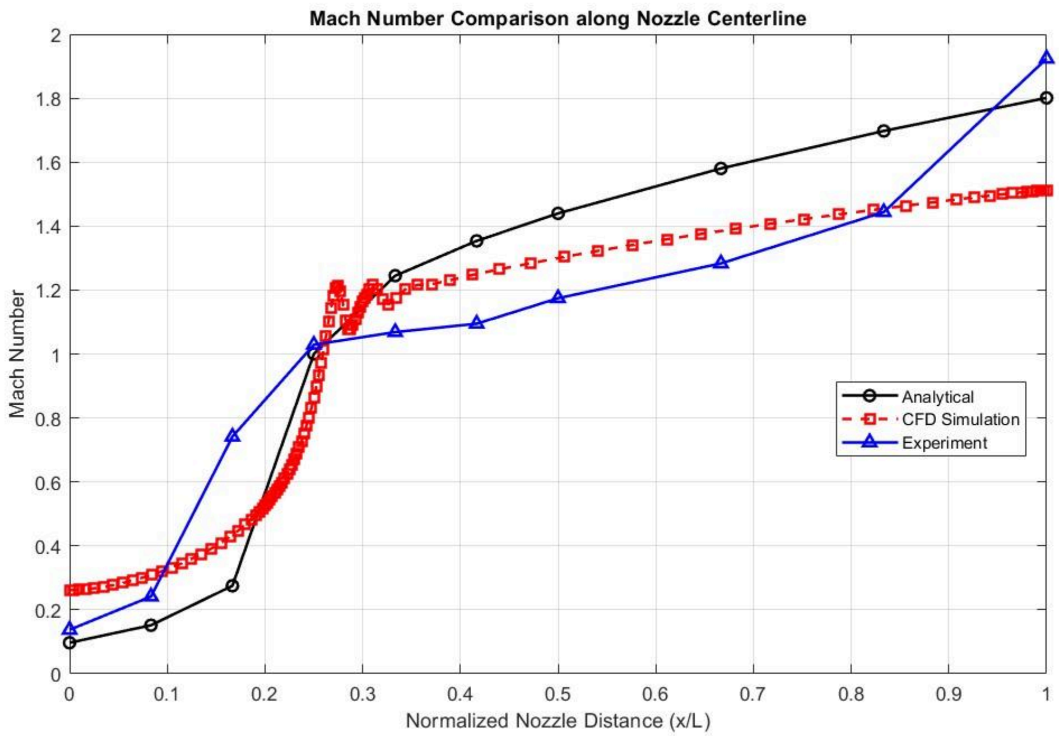

Figure 7: Mach number comparison (Analytical, CFD, and experimental)

In fig (8) The experimental results were converted into SI units, normalized and compared next to the normalized analytical isentropic flow predictions and the CFD simulations. The mach number showed a typical increase through a diverging section which confirms the transition to supersonic flow under specific choked conditions, with the flow reaching Mach 1 at the throat approximately 25% through the nozzle

**D.	Stagnation temperature along centerline**

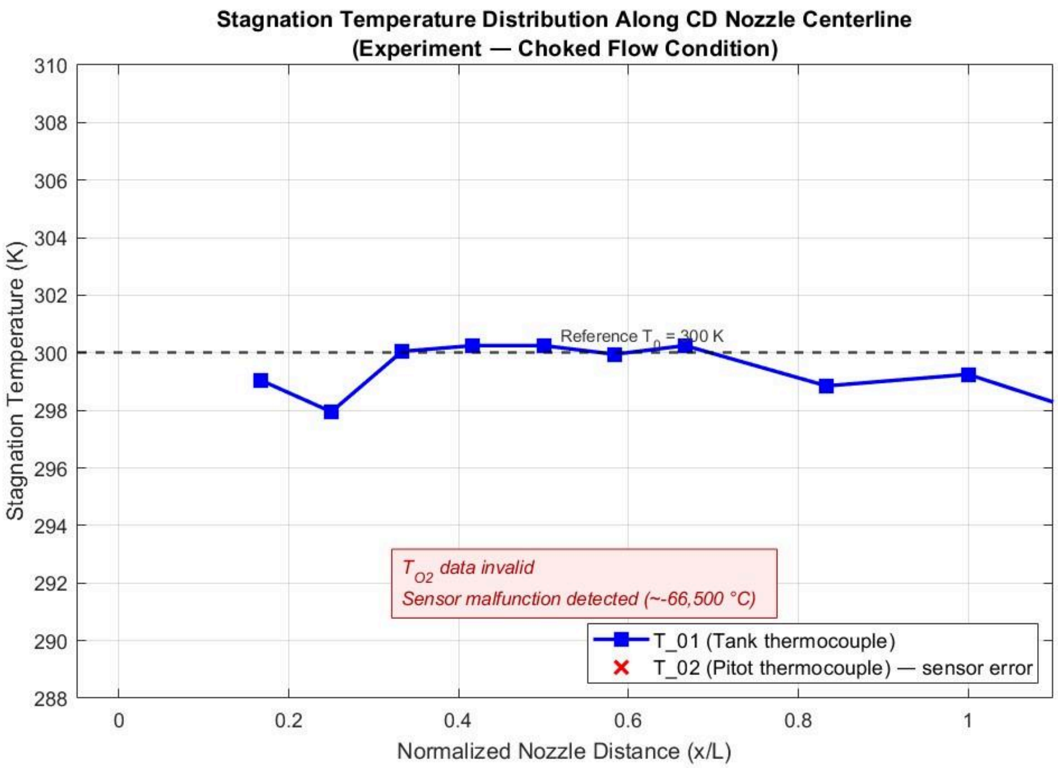

Figure 9: Stagnation Temperature Distribution

The reservoir temperature (T01) remained fairly stable near the reference values of 300K. The Pitot thermocouple values were excluded due to a sensor error that reported non possible values of -66,500C. All of these findings validate the 1D isentropic theory for CD nozzle design.

IV. Conclusion

The project successfully showed the process of designing, 3-D printing, and testing a CD nozzle. Through analytical calculations we found that the nozzle reached choked conditions at the throat which provided us a baseline for comparison. CFD simulations further drove us to more complex values and flow geometries that we wouldn't be able to calculate. The coplot of Mach number and stagnation temperature distributions revealed the CFD simulation is similar in the converging section, but lower than the diverging section, and the experimental flow is even higher in the converging and lower than the diverging. The experimental temperature results are inconclusive due to a faulty thermocouple reading from the pitot tube.

The source of the difference in the experimental, simulated, and analytical mach number is likely due to a combination of factors: the ignored boundary layer and the presence of the pitot tube. The analytical analysis assumes the flow is frictionless and inviscid, whereas the flow would actually choke slightly more than the geometry would suggest given the presence of a boundary layer. The CFD simulation more accurately models friction and boundary layers, leading to it being slightly less than the analytical Mach number. The CFD, however, does not consider the pitot tube used in the experiment. The pitot tube, in order to take measurements, also blocks the flow and constricts it further. This change in flow geometry leads to the flow choking more aggressively than the nozzle geometry would suggest, or that a simulation without a pitot tube would suggest. Because the flow is overexpanded, there is an oblique shock at the nozzle exit causing a pressure drop, which when plugged into Eq (3) calculates a higher Mach number at the exit. This explains why the experimental mach number was higher than predicted.

This experiment demonstrates the usefulness of analytical analysis and simulations to nozzle design. They enable a nozzle design to be tested without the resources needed to actually print and physically test the nozzle. Analytical methods are much simpler and faster to calculate, but simulations allow for more accurate and complex results, even with more complex nozzle designs. In general the project was widely a success and provided a valuable insight into the way compressible flow behaves and the uses in aerodynamic testing.

Appendix 1: 3D Printed Nozzle

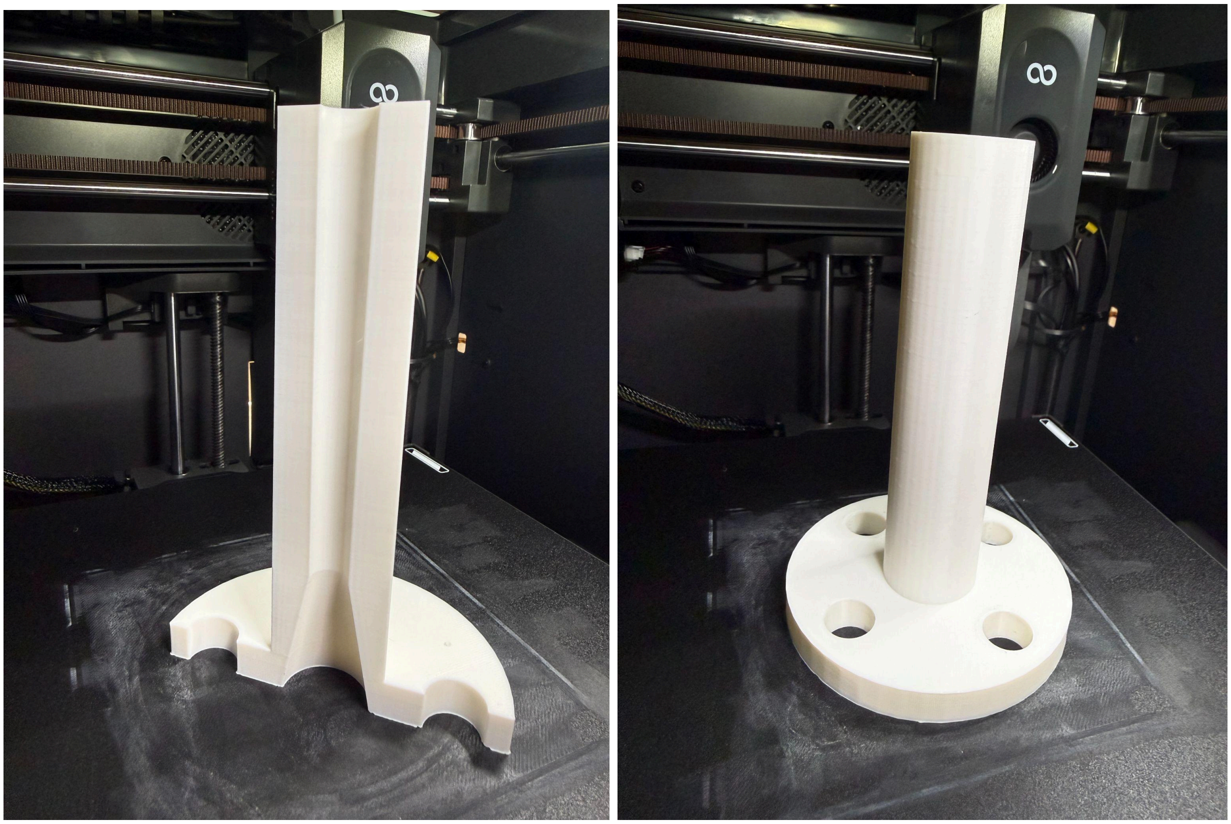

Figure 10: Finished 3d printed nozzle

The nozzle was printed out of PLA+ filament on a Centauri Carbon with a 0.4mm nozzle.

Appendix 2: Computational Mesh

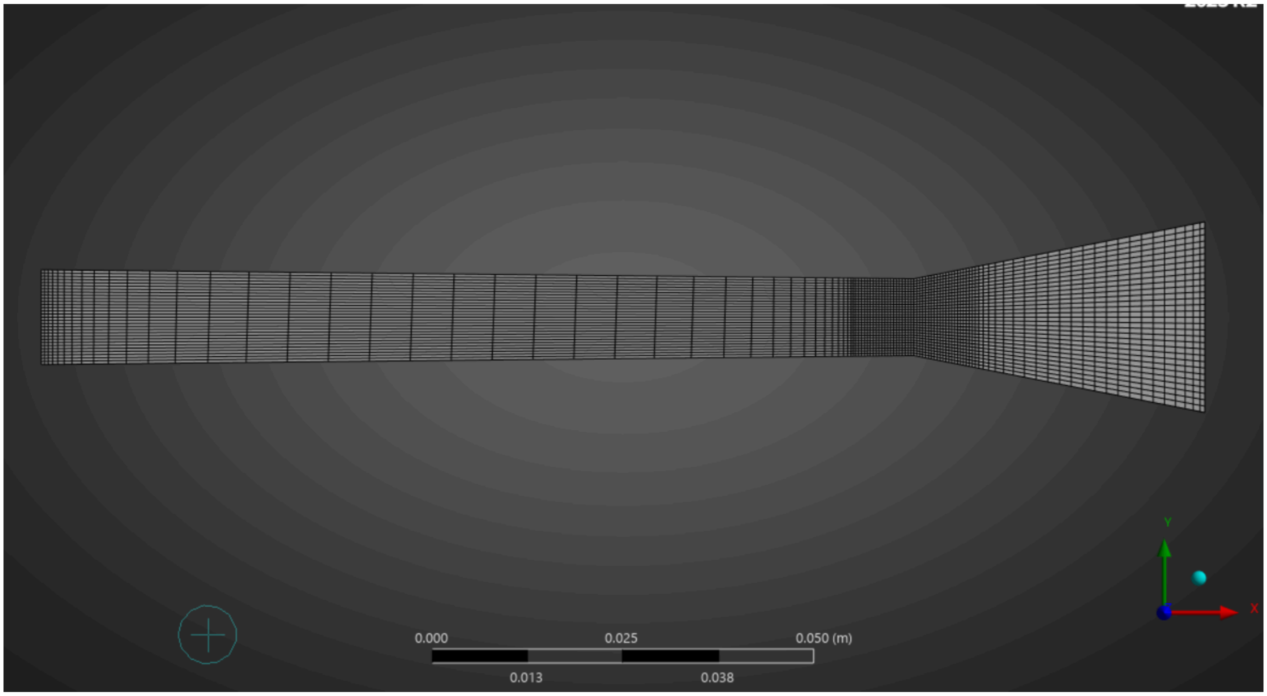

Figure 10: Ansys Fluid Flow Meshing

The meshing sequence was completed by following the lab 5 handout video tutorial. A quadrilateral dominant method and face meshing was used to get an accurate and detailed mesh of the CD geometry.

Appendix 3: Matlab Code

```
clear all; clc
gamma= 1.4;
R= 287;
P_0= 482633;
T_0= 300;
rho_0= P_0 / (R*T_0);
x= [0,.5,1,1.5,2,2.5,3,4,5,6,];
r=zeros(1,length(x));
for i=1:length(x)
   if x(i) <= 1.5
       r(i)= (.49- x(i)*tan(atan( (.49-.2)/1.5 )));
   else
       r(i)= (.2+ (x(i)-1.5)*tan(atan( (.24-.2)/4.5 )) );
   end
end
r= r.*25.4;
A= pi .* (r.^2);
xmm= x.*25.4;
AreaEq= @(M_val, A_rat) ((gamma + 1)/2)^(-((gamma+1) / (2*(gamma-1))) )
* ( ( (1 + ((gamma - 1)/2) * M_val^2)^ ((gamma+1) / (2*(gamma-1)) )
/M_val )) - A_rat;
A_star= A(4);
M= zeros(1,length(x));
for i = 1:length(x)
   A_rat = A(i) / A_star;
   if A_rat == 1
       M(i)= 1;
   elseif i <= 4
       M(i)= fzero(@(M_val) AreaEq(M_val, A_rat), .5);
   else
       M(i)= fzero(@(M_val) AreaEq(M_val, A_rat), 2.5);
   end
end
M=transpose(M);
%pressure
P= P_0 .* (1 + ((gamma -1)/2).*M.^2) .^ -(gamma/(gamma-1));
P=transpose(P);
%Temperature
T= T_0 .* (1 + ((gamma -1)/2).*M.^2) .^ -(1);
T=transpose(T);
%density
rho= rho_0 .* (1 + ((gamma -1)/2).*M.^2) .^ -(1/(gamma - 1));
rho=transpose(rho);
%velocity
a= sqrt(gamma*R.*T);
a=transpose(a);
```

```
v= M .* a;
v=transpose(v);
%mass flow rate
m_dot= (rho.*A.*v)/10^6;
P_psi= P./6894.76;
gamma = 1.4;
R = 287;
L_nozzle = 6.00 * 0.0254;
x_loc = ([7.00, 6.00, 5.00, 4.00, 3.50, 3.00, 2.50, 2.00, 1.50, 1.00]
-1) * 0.0254;
x_exp = x_loc / L_nozzle;
P02_psi = [67.3, 78.2, 86.2, 86.6, 85.4, 84.8, 85.4, 83.7, 83.0, 83.8];
Ps_psi  = [12.8, 24.4, 32.4, 37.1, 40.3, 41.3, 43.6, 58.1, 79.7, 82.7];
P02 = P02_psi * 6894.76;
Ps  = Ps_psi  * 6894.76;
T01_C = [24.5, 26.1, 25.7, 27.1, 26.8, 27.1, 27.1, 26.9, 24.8, 25.9];
To1_K = T01_C + 273.15;
M_exp = sqrt(( (P02 ./ Ps).^((gamma-1)/gamma) - 1 ) * (2 / (gamma-1)));
x_na= x/6;
%mach calc
RayleighEq = @(M_val, P_rat) ((gamma + 1) / (1 - gamma + 2 * gamma *
M_val^2)) * (((gamma + 1)^2 * M_val^2) / (4 * gamma * M_val^2 - 2 *
(gamma - 1)))^(-gamma / (gamma - 1)) - P_rat;
P_rat= Ps./P02;
M_e= zeros(1,10);
for i=1:10
   if P_rat(i) > .528
       M_e(i) = sqrt( (2 / (gamma - 1)) .* ( (P_rat(i)).^(-(gamma - 1) /
gamma) - 1 ) );
   else
       M_e(i) = fzero(@(M_val) RayleighEq(M_val, P_rat(i)), 2.5);
   end
end
CFD= readmatrix("Ansys.xlsx");
M_CFD= CFD(:,2);
x_CFD= CFD(:,1);
x_nCFD=x_CFD/6;
%Plot 1: Mach Number Comparison ---
figure('Name','Mach Number Comparison','Position', [100 100 900 560]);
hold on; grid on; box on;
plot(x_na, M, 'k-o', 'LineWidth', 1.5, 'DisplayName', 'Analytical');
plot(x_nCFD, M_CFD, 'r--s', 'LineWidth', 1.5, 'DisplayName', 'CFD
Simulation');
plot(x_exp, M_e, 'b-^', 'LineWidth', 1.5, 'DisplayName', 'Experiment');
xlabel('Normalized Nozzle Distance (x/L)');
ylabel('Mach Number');
title('Mach Number Comparison along Nozzle Centerline');
legend('Location','northeast');
```

```
%Plot 2: Stagnation Temperature
figure('Name','Stagnation Temperature','Position',[150 150 900 560]);
hold on; grid on; box on;
plot(x_exp, To1_K, 'b-s', 'LineWidth', 2.0, 'MarkerSize', 8, ...
   'MarkerFaceColor','b', 'DisplayName', 'T_01 (Tank thermocouple)');
% dummy entry for t02
plot(nan, nan, 'rx', 'MarkerSize', 10, 'LineWidth', 2, ...
   'DisplayName', 'T_02 (Pitot thermocouple) – sensor error');
yline(300.0, 'k--', 'LineWidth', 1.5, 'HandleVisibility','off');
text(0.52, 300.5, 'Reference T_0 = 300 K', 'FontSize', 9, 'Color',[0.2
0.2 0.2]);
annotation('textbox', [0.38 0.18 0.38 0.12], ...
  'String', {'\itT_{02} data invalid', 'Sensor malfunction detected
(~-66,500 °C)'}, ...
  'FitBoxToText','on', 'BackgroundColor',[1 0.93 0.93], ...
  'EdgeColor',[0.8 0 0], 'Color',[0.7 0 0], 'FontSize', 10);
xlabel('Normalized Nozzle Distance (x/L)', 'FontSize', 13);
ylabel('Stagnation Temperature (K)',        'FontSize', 13);
title({'Stagnation Temperature Distribution Along CD Nozzle Centerline',
...
     '(Experiment – Choked Flow Condition)'}, 'FontSize', 14,
'FontWeight','bold');
legend('Interpreter','latex', 'Location','southeast', 'FontSize', 11);
xlim([-0.05, 1.1]);
ylim([288, 310]);
set(gca, 'FontSize', 11);
```